

Standard Central Difference: $f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$

$$\text{Error} = \frac{h^2}{6} f''(x) + O(h^4)$$

①

Central Difference Basics

• approximates the derivative of a function by using values symmetrically around a point.

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h} \quad \left. \begin{array}{l} \text{First-order derivative approximation} \\ \text{w/ 2nd-order accuracy} \\ \text{error } O(h^2) \end{array} \right\}$$

$h = \text{step size}$

Central difference formula is derived by assuming the function can be approximated by quadratic polynomial (via Taylor series expansion) & the error terms of order h^2 & higher are neglected.

$$\text{2nd Derivative } f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

Fits quadratic curve through 3 points $(x-h, x, x+h)$

Error Order:

- ① Write the Taylor Series expansion of the function at points used in the approximation.
- ② Sub these into the finite difference formula
- ③ Simplify; remaining values is the leading error term.

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2} f''(x) + \frac{h^3}{6} f'''(x) + O(h^4)$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2} f''(x) - \frac{h^3}{6} f'''(x) + O(h^4)$$

$$f(x+h) - f(x-h) = 2hf'(x) + \frac{h^3}{3} f'''(x) + O(h^5)$$

↓
divide by $2h$

$$\text{Error} = \frac{h^2}{6} f'''(x) + O(h^4)$$

Second-order accuracy makes it more precise than forward/backward differences which are first-order.